

AIR UNIT #26 — THERMAL

ID:

obs-air-thermal

Role: High-Resolution Heat Detection / Night Recon

Tier: Recon Support

Status: Active

● **PHYSICAL & TECHNICAL**

Length: 2.1 m

Width: 1.5 m

Height: 0.8 m

Mass: 15 kg

Power Source: Mid-capacity battery

Energy Capacity: 1700 Wh

Energy Consumption: 36 Wh/min

Endurance: 35–55 min

Payload Capacity: 1.0 kg

Modules: High-resolution thermal camera

Navigation System: GPS + Thermal AI

Structural Integrity: 110 HP

Armor Rating: 0.14

Heat Signature: Very Low

Noise Signature: Very Low

Top Speed: 19 m/s

Acceleration: Medium

Turning Radius: Tight

Climb Rate: Medium

Stability Rating: High

● **AUTONOMY & AI BEHAVIOR**

Behavior Profile: Thermal Recon / Heat Detection

Aggression: 0.0

Caution: 0.9

Persistence: 0.85

Flanking Likelihood: 0.0

Retreat Threshold: 0.9

Target Priority: Heat signatures / movement

Formation Discipline: Medium

Risk Tolerance: Low

Sensors

Visual Range: 1400 m

Thermal Range: 2000 m (very high)

Acoustic Range: Low

Radar Range: None

LiDAR Range: Short-range

Signal Detection Range: Medium

Communication

Mesh Range: 5 km

Relay Capacity: 0

Command Link Type: Recon Grid

Network Priority: Medium

COMBAT & THREAT MODEL

Firepower Rating: 0

Accuracy: N/A

Stabilization: High

Weapon Mounts: None

Weapon Types: None

Suppression Value: 0

Tracking Strength: High

Disruption Strength: Low

Breach Capability: None

Countermeasure Rating: Low

Threat Modeling

Threat Level: 2

Engagement Distance: N/A

Preferred Terrain: Low-light airspace

Preferred Targets: None

Vulnerability Profile: Ballistic / weather

● SQUAD & COMMAND INTEGRATION

Squad Role: Thermal Recon

Squad Size Contribution: 1

Formation Behavior: Night-ops spacing

Intent Interpretation Rules: Maintain thermal visibility

Retreat Behavior: Early

Breach Behavior: None

Flanking Behavior: None

Reinforcement Behavior: Supports recon and command squads

Command Hierarchy

Parent Command Unit: Overseer / Insight

Subordinate Units: None

Relay Nodes: None

Network Dependencies: Recon Grid

● VR OPERATOR MODE

Operator Seat Type: Drone View

Operator HUD: Thermal Recon Overlay

Camera Feeds: Optical

Thermal View: Yes (high-resolution)

LiDAR View: Yes (short-range)

Sonar View: No

Tactical Overlay: Heat markers / movement indicators

Operator Controls

Movement Controls: Full drone flight

Sensor Controls: Thermal / LiDAR / zoom

Ability Controls: Thermal Scan, Mark Target, Stabilize

Weapon Controls: None

Stabilization Controls: Auto-hover

Operator Abilities

High-Resolution Thermal Scan

Mark Heat Target

LiDAR Sweep

Relay Recon Data

● ECONOMY & LOGISTICS

Build Cost: 100 Metal / 90 Electronics

Build Time: 30 sec

Factory Type: Drone Bay

Tech Requirements: Recon Systems II

Maintenance Cost: Low

Repair Time: 10 sec

Supply Consumption: Low

Operational Cost: Minimal

Deployment Requirements: Standard drone pad

Transport Compatibility: All aerial racks

Launch Method: Vertical takeoff

🔗 BUILD REQUIREMENTS (PLAYER COLLECTION SYSTEM)

Common Components

10× Light Alloy Plates

6× Composite Frame Rods

4× Micro-Servo Actuators

2× Stabilizer Fins

1× Thermal Sensor Housing

Electronics

10× Circuit Bundles

8× Signal Processors

3× Drone Control Chips

2× Thermal AI Chips

Power

1× Mid-Capacity Battery Pack

2× Power Regulation Nodes

Rare / Tradeable

1× High-Resolution Thermal Core

Optional Upgrades

Silent Rotor Kit

Extended Battery Pack

Ultra-Thermal Lens